

Formal Report on Practical Training CON2013X

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TERMS OF REFERENCE

This report has been prepared to fulfil the practical training requirements of CON2013X as prescribed by the Department of Construction Economics and Management at the University of Cape Town.

The report reflects on practical training undertaken with H&I Construction, a privately owned multi-disciplinary South African construction company operating across a range of sectors including roads and earthworks, civil and building construction, water infrastructure, mechanical and electrical works, and other major engineering projects. H&I has operated nationally since 1984 and delivers projects for both public and private sector clients.

The practical training was completed over two vacation periods between 23 June 2025 and 10 June 2026 and amounted to a total of 229 hours of supervised work experience. Although H&I operates across numerous construction disciplines, the students placement was specifically within a project management team responsible for administering City of Cape Town maintenance and infrastructure contracts under the 34Q and 144Q term tenders.

The training was conducted primarily at Tygerberg Site Camp 2 and later at H&I's Durbanville Head Office. During this period, exposure was gained to project management, contract administration, quantity surveying, procurement processes, document control, project planning and site inspections. The practical training also provided opportunities to interact with project managers, quantity surveyors, engineers, site agents and other construction professionals involved in the delivery of municipal projects.

The purpose of this report is to document the practical experience obtained, reflect on the lessons learned, evaluate the development of professional and technical skills, and demonstrate how the experience contributed towards a broader understanding of the construction industry and the learning outcomes of the BSc Construction Studies programme.

SUMMARY

This report provides a reflection on the practical training undertaken with H&I Construction between 23 June 2025 and 10 June 2026, totalling 229 hours of supervised practical experience. The training was completed as part of the requirements for CON2013X across 3 vacation periods and provided the student with exposure to the managerial, contractual and administrative aspects of the construction industry.

The practical training was conducted at Tygerberg Site Camp 2 and later at H&I's Durbanville Head Office. Throughout the training period, the student worked alongside project managers, quantity surveyors, engineers and other construction professionals involved in the management and delivery of City of Cape Town's maintenance and infrastructure projects under the 34Q and 144Q term tenders.

The students responsibilities included assisting with workload planners, processing Purchase Orders, reviewing subcontractor quotations, compiling project documentation, preparing Purchase Order Alerts and Letters of Appointment. The student also undertook quantity measurements using drawings and digital measurement software, and assisting with project administration activities. In addition, the student accompanied project managers and quantity surveyors on various site inspections where defects, workmanship standards, snag lists and practical completion requirements were assessed.

One of the most significant aspects of the students training was how the student gained an understanding of how construction projects are managed beyond the physical work taking place on site. The experience highlighted the importance of contract administration, procurement processes, document control, financial management and communication in ensuring successful project delivery. It also demonstrated the role that quantity surveyors and project managers play in coordinating contractors, managing project costs and ensuring contractual obligations are met.

The practical training provided an opportunity to apply concepts learned through Construction Studies courses such as Construction Management, Measurement and Design Appraisal, and Business Law. The training also contributed significantly to professional development by improving the students confidence, communication skills, attention to detail and understanding of the construction industry.

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1. INTRODUCTION

1.1. Subject of and Motivation for Report

This report documents the practical training undertaken by Sage Agnew, a fourth-year BSc Construction Studies student at the University of Cape Town, during employment with H&I Construction in 2025 to 2026. The motivation for this report stems from the degree requirement to complete 480 hours of supervised practical experience within a built environment discipline. The report aims to demonstrate the integration of theoretical knowledge acquired throughout the students degree and to gain exposure of the practical realities of a professional construction environment.

1.2. Background to Report

This report has been prepared to fulfil the practical training requirements of CON2013X as prescribed by the Department of Construction Economics and Management at the University of Cape Town.

The report reflects on practical training undertaken with H&I Construction, a privately owned, multi-disciplinary South African construction company operating across a range of sectors. The practical training was completed over two vacation periods between 23 June 2025 and 10 June 2026, amounting to a total of 229 hours of supervised work experience.

The placement was primarily within a project management team responsible for administering City of Cape Town maintenance and infrastructure contracts under the 34Q and 144Q term tenders. Training was conducted primarily at Tygerberg Site Camp 2 and later at H&I's Durbanville Head Office. During this period, I was exposed to project management, contract administration, quantity surveying, procurement, document control, project planning and site inspections.

The purpose of this report is to document my practical experience, reflect on the lessons learned, evaluate my development of professional and technical skills, and demonstrate how the experience contributed to my broader understanding of the construction industry and the learning outcomes of the BSc Construction Studies programme.

1.3 Objectives of Report

The objectives of this report are to:

- Document the daily activities and tasks undertaken during the practical training period.
- Provide a reflective analysis of the skills acquired, linking them to the BSc Construction Studies degree outcomes.
- Fulfil the assessment requirements for the course CON2013X.

1.4. Limitations and Scope of Report

The practical training consisted of 229 hours of supervised work experience completed over two vacation periods. The scope of the placement included project administration, workload planning, Purchase Order processing, contract documentation, quantity measurements, subcontractor quotation reviews, site inspections and quality control activities.

A limitation of the practical training was the relatively short duration of the vacation periods. Construction projects often span several months or years, making it difficult to observe every stage of the project lifecycle during a single training period. As a result, exposure was limited to specific project phases and activities taking place during the time of the placement. Despite this limitation, the variety of projects and responsibilities encountered provided valuable insight into the construction industry and the role of contractors within the built environment.

1.5. Plan of Development

The report begins with an overview of the practical training experience before presenting a reflective discussion of the lessons the student learned as well as technical observations and professional development achieved throughout the placement. Particular attention is given to project management, contract administration, quantity surveying, site inspections and the relationship between theoretical knowledge and industry practice.

1.6. Research Approach

The report is based primarily on observations made during the practical training period, discussions with construction professionals, project documentation reviewed during the placement and personal reflections recorded in the daily logbook. Additional information was obtained through company documentation, construction drawings, Bills of Materials, workload planners and contractual records encountered during the training period.

2. REFLECTIVE ESSAY

2.1 Company Background

H&I Construction, originally established as Haw & Inglis Civil Engineering in 1984, is a privately owned South African construction company headquartered in Cape Town. Since its establishment, the company has expanded from its original civil engineering focus into a multi-disciplinary contractor operating across South Africa and selected Southern African countries.

The company undertakes projects across a range of sectors, including road construction and rehabilitation, structural and bridge works, water and wastewater infrastructure, marine and port-related projects, renewable energy, and commercial, industrial and residential developments. This broad range of activities reflects the diverse nature of the construction industry and provided the wider organisational context for my practical training.

My placement exposed me primarily to the project management, quantity surveying and contract administration functions of the business, providing an opportunity to observe how the financial, contractual, technical and administrative aspects of construction are coordinated within a professional contracting environment.

2.2 Beginning of Practical Training

My opportunity to become part of H&I Construction began in 2025 when I was awarded a study bursary through the company. Coming from a background where continuing my university studies was not something I could take for granted, receiving this support was significant to me. The bursary provided financial assistance towards my studies, but its value extended beyond financial support. As part of the study sponsorship agreement, I was also required to complete practical training within the company during my vacation periods. This aligned directly with the practical training requirements of the BSc Construction Studies degree at UCT.

This created an opportunity that was particularly meaningful because I was able to fulfil two important requirements simultaneously where I could gain the practical experience required by my degree while also meeting the training requirements of my study sponsorship. Instead of having to search independently for a placement, I entered an environment where the company had already invested in my education and had an interest in my development as a future construction professional.

Knowing that I had been given an opportunity within a company that had supported my studies gave me a sense of security when beginning my training. At the same time, it created a sense of responsibility because I wanted to make the most of the opportunity that had been given to me. I was not entering H&I simply to complete a university requirement, I was entering an organisation that had invested in my development and, through the sponsorship agreement, had an interest in seeing me grow professionally.

This made the beginning of my practical training different from my previous construction experience. I approached the placement with a combination of excitement, gratitude and pressure to perform. I knew that I had much to learn, but I also felt that I had been given a valuable opportunity to develop within a company that wanted to see me succeed. In my mind, this provided me with a comfortable environment in which to learn, make mistakes, ask questions and gradually develop my confidence. The practical training therefore represented more than an academic requirement, it actually was an early opportunity to begin building a professional relationship with a company and to understand what would be expected of me within the construction industry.

2.3 Reflective essay

My practical training at H&I Construction significantly changed my understanding of the construction industry. Before joining H&I, my experience had mainly been site-based through Feytcon Construction, where I worked as a labourer on paving, bricklaying, earthworks and related activities. I therefore expected H&I to involve primarily site observation and physical construction. Instead, much of my training took place in an office alongside project managers, quantity surveyors and engineers. This challenged my assumption that construction is defined mainly by what happens on site. I came to understand that successful project delivery also depends on planning, procurement, financial management, contract administration and communication. This broader perspective became the foundation of my learning during the placement and further reinforced the concepts taught in class as I got reflect on the planning, procurement and financial aspect I learnt through my coursework.

One of my biggest adjustments was adapting to the professional office environment. At first, I felt pressure to perform well because I believed that, as a UCT Construction Studies student and H&I bursary recipient, I should already be competent. I often tried to complete tasks independently and quickly because I did not want to appear inexperienced. However, this sometimes resulted in mistakes or unnecessary delays because I was reluctant to ask questions. I gradually realised that practical training is

intended to expose students to unfamiliar situations and not test whether they already know everything. My colleagues encouraged me to ask for clarification, and as I became more comfortable doing so, my confidence and quality of work improved. By the end of the placement, I was actively requesting additional work and seeking feedback rather than avoiding questions. This taught me that professional competence is not simply knowing the answers, it also involves recognising when to ask for assistance and being willing to learn.

A major area of learning was my exposure to the commercial and administrative side of construction. I worked within a team administering City of Cape Town projects under the 34Q and 144Q term tenders. Concepts such as procurement, Bills of Quantities (BOQs), tender rates and Purchase Orders had previously been largely theoretical to me in class. At H&I, I could see how they operated within live projects. I learned that the 34Q contract generally involved heritage building maintenance, while 144Q covered water and sanitation-related additions, alterations, repairs and maintenance. Understanding these contracts helped me see how construction work is organised within a municipal environment and how contractual requirements influence project delivery.

Working with BOQs and subcontractor quotations was particularly valuable. I compared subcontractor rates with H&I's tendered rates and observed how project managers and quantity surveyors considered whether work packages were commercially viable before appointing subcontractors. This demonstrated that the financial success of a project can be influenced well before work begins on site. When I was tasked to do these comparisons I was basically comparing the rates of us as the contractor to that of the subcontractor and I would highlight the relevant BOQ item that was causing a huge discrepancy or if it would cause us to make a loss. I would then bring this to the attention of the senior Quantity surveyor in office to make him aware and then I book this and organise the rest of the comparisons I made into a file.

Similarly, processing Purchase Orders showed me the connection between procurement, funding and project execution. I assisted with recording Purchase Order numbers, dates, durations and scopes, where I updated project statuses after talking to the relevant project manager in office and then I would attach the supporting documentation such as Purchase Order Alerts and Letters of Appointment. Previously my first encounter with Purchase Orders were while organising residence events at UCT, but here it was so much more different seeing them used on construction projects and I can confirm it demonstrated much greater contractual and financial significance.

While I was here I also assisted with maintaining workload planners used by project managers. These contained project information, Purchase Orders, quotations, Letters of Appointment for subcontractors and other contractual documentation. Initially, the volume of information was overwhelming, and I struggled to understand which documents were relevant. I had to go back and forth taking all the different Project managers and quantity surveyors involved with the different projects inside these workload planners where what goes where and how I should organise everything but this was actually teaching me something greater because as I began getting the hang of organising each project I could start understanding the procedure of how things had to

unfold from start all the way to final completion. However, working through these records taught me the importance of document control and accurate information management. I also learned that seemingly small administrative errors can affect project records, procurement and financial decisions. This reinforced the importance of attention to detail, particularly in an industry where professionals make decisions based on information recorded by others.

With that being said there were so many errors that I picked up and I was proud to have picked things up like whether or not a project still needed to be looked into or if it can be booked as finally complete.

My exposure to quantity surveying further strengthened this understanding. I worked alongside Senior Quantity Surveyors Ieliyaas Abderoef and Rodger Phillips and observed how they measured quantities, reviewed costs and monitored financial aspects of projects. I was later introduced to digital measurement software and undertook quantity take-offs from construction drawings, including masonry, plasterwork, paintwork, concrete and structural steel. Initially, one of the Junior Quantity surveyors tasked me to measure a drawing and without the necessary software he helped me get around this by introducing me to a software called Methvin where I was able to measure a drawing seamlessly, it was online and I found the software and measurement procedures difficult at first but the more time I spent I using it I quickly became used to it, but I began to appreciate the systematic nature of quantity measurement. I was also taught how to store my measurements in excel the “old school way” where I would later start Comparing my measurements with those completed by a junior quantity surveyor. He also later showed me how easily rushing could introduce inaccuracies. This connected directly with a course I had been doing called Measurement and Design Appraisal I, where I had learned that accurate measurement forms the basis of estimating and cost planning. Additionally we would later compare these measurements on site using a tape measure to make sure the work being done on site was accurate. At H&I, I saw that these quantities could ultimately influence subcontractor payments and project budgets.

The City of Cape Town financial year-end provided another important perspective. This seemed to introduce a time of stress and urgency because during my first weeks at H&I, project teams were under pressure to achieve Practical Completion before the end of June. This was the end of the city of Cape Town’s financial year. I initially saw this as simply a deadline, but I came to understand that contractual and financial requirements also influence project programmes. I observed increased communication, more site inspections and efforts to obtain approvals for practical completion and sign-offs. This connected with concepts from Construction Management 1 taught in second year regarding programme control, stakeholder coordination and Practical Completion, while demonstrating that project deadlines are shaped by more than physical progress alone.

Although much of my training was office-based, site inspections were among the most valuable experiences because they allowed me to see the consequences of decisions made in the office. At Wallacedene in Kraaifontein, I accompanied Project Manager Alec Kanyerere to inspect a steel-framed structure. We identified poor welding, skew

structural elements and poorly finished concrete column pads. I definitely overlooked some of these defects because of my limited experience, and because I didn't know if I had the capacity to even mention if something seemed wrong but the project manager identified them immediately. This made me realise that construction quality depends not only on specifications and checklists but also on experience and a trained eye. It motivated me to develop the technical knowledge required to recognise problems for myself.

Another particularly significant experience was attending a Practical Completion inspection at the Philippi East Fresh Produce Market, where representatives of the City of Cape Town, the subcontractor, as well as a facility management and a candidate Professional Quantity Surveyor discussed outstanding defects and whether they had been adequately addressed. I initially expected a straightforward inspection, but the meeting was more complex. The project manager from H&I had to balance the client's expectations, the contractor's constraints and the subcontractor's obligations while remaining guided by the contract. Discussions regarding galvanisation, fencing and repair methods showed me that project management involves continual negotiation and communication between stakeholders. This was my first meaningful observation of the JBCC being applied in practice. I also gained a clearer understanding of Practical Completion as a contractual milestone rather than simply the point at which construction appears finished. Work must meet the required standard and outstanding defects must be addressed before the facility can be accepted as practically complete.

Additionally At Belroad Fire Station, a separate project completed through H&I, I was responsible for photographing completed subcontractor work for the project manager's reports. Although this task was relatively simple, I approached it differently because of what I had learned from previous inspections. Here, Instead of merely taking photographs, I actively looked for potential defects and considered what the project manager would need to demonstrate completed work. Being trusted with this responsibility gave me a sense of professional confidence and showed me how even small tasks contribute to project administration.

These experiences also strengthened the connection between my university studies and professional practice. Construction Management helped me understand programme control, stakeholder relationships and project coordination. Measurement and Design Appraisal supported my understanding of quantity take-offs, while Business Law 1 became more meaningful through exposure to contracts, procurement, BOQs and subcontractor quotations. Concepts that had previously existed mainly in lecture notes became practical processes with financial and contractual consequences. This helped me appreciate why the theoretical foundations of my degree are important.

The most significant change, however, was personal. I began the placement believing that I needed to prove that I was capable, but I finished understanding that learning requires vulnerability and communication. I became less afraid of making mistakes and more willing to ask questions, accept correction and seek feedback. If I were to repeat the placement, I would ask more questions earlier, seek exposure to different departments and continue developing my understanding of contract administration and

digital construction tools. Most importantly, I would avoid placing unnecessary pressure on myself to know everything before beginning a task.

Overall, H&I broadened my understanding of construction from physical site activities to a profession involving technical, commercial, contractual and interpersonal responsibilities. The experience confirmed my interest in construction management while also exposing me to the contractor quantity surveying and contract administration. More importantly, it taught me that professional development is continuous. Technical knowledge is essential, but so are accuracy, communication, adaptability into the environment you're in and the willingness to learn from others. These lessons have provided a stronger foundation for my continued development as a construction professional.

3. CONCLUSION

My practical training at H&I Construction transformed my understanding of how construction projects are delivered. Moving from previous labour-based site experience into an environment involving project management, quantity surveying, procurement and contract administration showed me that successful construction depends on much more than physical work. Working with BOQs, Purchase Orders, workload planners and subcontractor quotations demonstrated the importance of accurate information and financial control, while site inspections and Practical Completion meetings showed me how contractual requirements influence the work ultimately delivered.

The placement also strengthened the relationship between my university studies and industry practice. Concepts from Construction Management, Measurement and Design Appraisal, Business Law and quantity surveying became more meaningful when applied to real projects. Most importantly, I developed greater confidence in my ability to learn. I entered H&I afraid that asking questions would expose my lack of knowledge however, left understanding that asking questions, accepting feedback and learning from mistakes are essential parts of professional development.

The experience has therefore not only expanded my technical understanding but also changed how I approach learning and responsibility. I leave the placement with a broader view of construction, a clearer understanding of the roles within the industry and an overall stronger foundation for my future career.

Reference List

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